debt awareness, and a preference for mobile notifications. This character represents:

A) A real user profile B) A user archetype / user persona
C) A market segment spreadsheet D) A stakeholder map

Q17. A designer uses an Empathy Map to analyze a banking user. In the 'Thinks and Feels' quadrant, the user's data shows "I feel like the bank is charging me hidder fees" even though the fees are clearly listed in the terms. This insight is crucial because it reveals:
A) An explicit user requirement.
B) A perceptual user pain point and emotional friction.
C) A functional bug in the code. D) A regulatory compliance issue.

Q18. A company creates a persona of a young professional but forgets to consider elderly users. Their product fails with the elderly segment. This highlights the importance of:
A) Focusing only on the largest segment
B) Creating multiple personas for diverse user groups
C) Reducing the number of features
D) Using only quantitative data

Q19. In an Empathy Map, the 'Says' quadrant shows "This app is so easy to use," but the 'Does' quadrant shows the user frequently gets stuck and asks for help. This discrepancy indicates:
A) The user is lying
B) A hidden pain point or unarticulated need
C) A software bug D) The user is distracted

Q20. Which quadrant of the Empathy Map captures what the user's friends, family, and media say that influences their decisions?

A) Thinks and Feels B) Says
C) Hears D) Pain Points

Q21. A team wants to know *what percentage* of users abandon their cart at each checkout step. Which research method should they use?
A) Qualitative Interviews B) Quantitative Analytics
C) Empathy Mapping D) Contextual Inquiry

Q22. IDEO's Human-Centered Design (HCD) framework organizes the creative process into which three core spaces?
A) Discover, Design, Deliver B) Empathize, Define, Test
C) Inspiration, Ideation, Implementation D) Observe, Orient, Execute

Q23. The primary purpose of the 'Prototype' phase in design thinking is to:
A) Launch the final product to the entire market.
B) Create a fully functional, production-ready version.
C) Make ideas tangible and test specific aspects of a solution quickly and cheaply.
D) Validate the business model before engineering begins.

Q24. Which of the following is an emotional pain point rather than a functional pain point?
A) The login page takes 8 seconds to load
B) The search bar doesn't show relevant results
C) I feel anxious when I see the price increase notification
D) The font size is too small to read comfortably

25. According to the VARK model, a student who learns best by taking detailed notes and rewriting them is primarily a _____ learner.
A) Visual B) Aural
C) Read/Write D) Kinesthetic

26. Which statement correctly distinguishes 'Invention' from 'Innovation'?
A) Invention is the refinement; Innovation is the initial discovery
B) Invention is the novel creation; Innovation is the scaling, refinement, and value extraction
C) Invention is always accidental; Innovation is always intentional
D) Invention and Innovation are synonymous

27. The Ford Edsel failed despite extensive market research. The Tata Nano failed because buyers wanted social status. The Segway failed because walking wasn't a problem. These failures collectively highlight the importance of:
A) More engineering investment
B) Understanding implicit emotional and psychological needs, not just explicit demands
C) Lowering production costs
D) Better marketing campaigns

28. A researcher observes a student: "Student (User) highlights notes (Activity), using a laptop (Object) in a library (Environment)." According to AEIOU, which element is missing from this observation?
A) Activities B) Environments
C) Objects D) Interactions

29. A productivity app detects that a user switches between tasks every 2 minutes and suggests a "Focus Mode" with timed intervals. This adaptive feature is an example of:
A) Functional fixedness
B) Computational empathy through behavioral analytics
C) Quantitative research D) Perceptual block

Q30. A researcher observes: "A shopper (User) reads product labels (Activity), uses a smartphone (Object) in a supermarket aisle (Environment), and asks the store assistant for help (Interaction)." Which element is the "supermarket aisle"?
A) Activity B) Environment
C) Object D) Interaction

Question	1	2	3	4	5	6
Answer						
Question	7	8	9	10	11	12
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A) Focused Mode only
B) Diffuse Mode processing allowing subconscious connections
C) Sensory overload inhibition
D) Analytical linear processing

Q2. A professor notices that some students understand concepts best through diagrams, others through lectures, and others through hands-on experiments. She decides to include all three methods in her teaching. This approach reflects which learning framework?

A) Bloom's Taxonomy
B) VARK Model
C) Maslow's Hierarchy
D) Piaget's Theory

Q3. A non-profit organization wants to build affordable housing. The team spends the first two weeks living in the local community, shadowing families to understand their daily routines, storage needs, and social habits. According to the Stanford d.school framework, which phase is the team currently executing?

A) Define
B) Ideate
C) Empathize
D) Prototype

Q4. After conducting 50 interviews, a product team synthesizes the data into a single actionable statement: "Young urban parents need a way to track their toddler's location securely in crowded public spaces without feeling overbearing." According to the Stanford d.school framework, they have just completed the _____ phase.

A) Empathize
B) Define
C) Ideate
D) Test

Q5. IDEO was tasked with redesigning a hospital cart. Instead of designing the final expensive electronic cart immediately, they built rough, low-fidelity prototypes using cardboard, wood, and PVC pipes to test the shape and height with nurses. In IDEO's Human-Centered Design framework, this rapid building to learn falls under which space?

A) Inspiration
B) Ideation
C) Implementation
D) Validation

Q6. A design team is stuck on generating diverse solutions. They use a technique where they set a 5-minute timer and sketch 8 radical ideas each. Which Stanford d.school phase and technique does this specifically belong to?

A) Empathize – User observation
B) Define – Finding insights
C) Ideate – "Crazy 8s" brainstorming
D) Test – Usability checking

Q6. A team designing a new school spends two weeks observing classroom interactions, talking to teachers, and understanding student behaviors. In the Stanford d.school framework, which phase is the team currently executing?

A) Define
B) Ideate
C) Empathize
D) Prototype

Q7. A product team synthesizes 75 user interviews into a single insight: "Remote workers need a tool that minimizes distractions while maintaining team connectivity." This statement is the output of which Stanford d.school phase?

A) Empathize
B) Define
C) Ideate
D) Test

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A) Empathize
B) Define
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D) Test

Q9. A team builds a Rough paper prototype of their product and asks users to walk through tasks. In IDEO's Human-Centered Design framework, this rapid building to learn falls under which space?

A) Inspiration
B) Ideation
C) Implementation
D) Validation

Q10. Which of the following scenarios best illustrates the "Implement" phase of IDEO's Human-Centered Design framework?

A) Conducting user interviews to understand pain points.
B) Brainstorming 100 different solutions to a problem.
C) Launching the product and creating a feedback loop for continuous improvement.
D) Building a low-fidelity paper prototype.

Q11. Airbnb's early success is partly attributed to the founders physically flying to New York, renting a professional camera, and taking high-quality photos of the hosts' apartments themselves. This action, which fundamentally increased user trust and bookings, is a classic example of:

A) Scale optimization.
B) Technical invention.
C) Getting out of the building (empathy and observation) to understand the real gap.
D) Reducing unit costs.

Q12. A streaming platform notices that a user frequently pauses, rewinds, and turns on subtitles when watching movies with heavy accents. The software automatically suggests a "Dialogue Boost" audio feature. This proactive adaptation to user behavior is best described as:

A) Quantitative user research.
B) Functional fixedness.
C) Computational empathy.
D) Perceptual block.

Q13. A navigation app detects that a user is driving erratically and gently suggests: "Would you like to take a break? There is a rest stop ahead." This proactive behavior-based response is an example of:

A) Functional fixedness
B) Computational empathy
C) Quantitative research
D) Perceptual block

Q14. A company wants to understand *why* users prefer a competitor's product. They conduct 1-hour, open-ended interviews with 30 users. This research is classified as:

A) Quantitative Research
B) Qualitative Research
C) Statistical Sampling
D) A/B Testing

Q15. A researcher surveys 20,000 customers to measure the exact percentage who are satisfied with a new feature. This approach is categorized as:

A) Qualitative Research
B) Generative Research
C) Quantitative Research
D) Exploratory Research

Q16. A credit card company wants to reduce customer churn. They create a fictional character named "Stressed Sam," a 35-year-old freelancer with irregular income, high

Name: [redacted]
Registration Number: [redacted]
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CA-1 (INT-335

Q1. A software developer spends 4 hours intensely debugging a piece of code. Frustrated, she steps away to take a shower. Suddenly, the solution to the bug pops into her head. Which brain processing dynamic best explains this sudden breakthrough?
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